

Resolved Mixed-Support and Structural Validation

JWST-SUSPENSE 129982 in Curvature Polarization Transport Gravity

Carter L Glass Jr

Abstract

JWST-SUSPENSE 129982 is a massive quiescent galaxy at $z = 1.249$ with unusually strong stellar rotation for its class and epoch. Slob et al. report $V_{R_e} = 334^{+52}_{-46} \text{ km s}^{-1}$, $\sigma_0 = 213^{+11}_{-2} \text{ km s}^{-1}$, $V_{R_e}/\sigma_0 = 1.57^{+0.26}_{-0.23}$, and $\lambda_{R_e} = 0.52^{+0.07}_{-0.06}$. The same source analysis gives a dynamical-to-stellar mass ratio $M_{\text{dyn}}/M_{\star} \simeq 4.79$ under its conventional thin-disc dynamical model. This combination of substantial ordered and random stellar motion makes 129982 a useful mixed-support gravity test and a valuable probe of the dynamical evolution of massive quiescent galaxies at cosmic noon.

Curvature Polarization Transport Gravity (CPTG) retains the measured baryonic distribution as the material source and represents the additional gravitational response geometrically through curvature polarization and transport rather than through an inserted dark-halo source term. Three antisymmetric inner rotation coordinates are reconstructed from the public JWST/NIRSpec two-dimensional spectrum, reaching approximately $0.62R_e$, while the published effective-radius rotation is withheld. An oblate Sérsic stellar source is then combined with the nonlinear CPTG field equation and a radial Jeans pressure boundary-value problem. The effective-radius pressure anchor selects $a_{\star} = 2.972 \times 10^{-9} \text{ m s}^{-2}$. The resolved-inner solution returns $V_{\text{CPTG}}(R_e) = 320.9 \text{ km s}^{-1}$, 13.1 km s^{-1} below the published central value and well inside its reported interval. The same solved field gives Structural Mode $N = 2.9191$, corresponding to the current Intermediate-Spiral CSMI interval. The result provides a resolved mixed-support CPTG cross-validation using one baryon-sourced field for both ordered and pressure support, followed by an independent post-solution measure of radial field organization.

1 Introduction: Mixed-Support Dynamics as a Gravity Test

JWST-SUSPENSE 129982 provides a rare combination of structural and dynamical measurements in a massive quiescent galaxy at cosmic noon. At $z = 1.249$, the system contains approximately $1.7 \times 10^{11} M_\odot$ in stars under a Chabrier initial mass function [3], has a source-reported effective radius $R_e = 6.48 \pm 1.13$ kpc, a concentrated Sérsic profile with $n = 3.6 \pm 0.1$, and a projected axis ratio $q = 0.39 \pm 0.01$ [1]. Despite its quiescent state, the galaxy retains substantial ordered stellar rotation.

The intrinsic effective-radius kinematics inferred by Slob et al. are

$$V_{R_e} = 334_{-46}^{+52} \text{ km s}^{-1}, \quad \sigma_0 = 213_{-2}^{+11} \text{ km s}^{-1}, \quad (1)$$

with

$$\frac{V_{R_e}}{\sigma_0} = 1.57_{-0.23}^{+0.26}, \quad \lambda_{R_e} = 0.52_{-0.06}^{+0.07}. \quad (2)$$

Among the JWST-SUSPENSE galaxies for which rotation is constrained, 129982 lies at the high end of the reported V/σ range [1]. This makes the object informative for two related reasons. Astrophysically, it helps trace the evolution of stellar angular momentum in massive quiescent galaxies. Dynamically, it occupies a regime in which neither circular rotation nor pressure support can be treated as a small correction to the other.

That mixed-support character is central to the present test. A viable gravitational description must support the measured stellar mass distribution, the rising ordered rotation, and the substantial random stellar motion within one consistent field solution. The source analysis also reports

$$\log_{10}(M_{\text{dyn}}/M_\odot) = 11.9 \pm 0.1, \quad \log_{10}(M_\star/M_\odot) = 11.22 \pm 0.01, \quad (3)$$

which gives

$$\frac{M_{\text{dyn}}}{M_\star} = 4.7863. \quad (4)$$

Under low-gas Chabrier-IMF bookkeeping, the source interpretation therefore requires a large non-stellar contribution to the dynamical budget. The same galaxy can consequently be used to contrast a conventional dark-matter accounting with a CPTG geometric-response calculation.

Curvature Polarization Transport Gravity (CPTG) is a geometric gravitational framework in which ordinary physical matter remains the material source. Additional gravitational response is carried by curvature-polarization and curvature-transport terms rather than by assigning an independent dark-sector stress-energy component to the local galaxy field [7]. In the galaxy reduction, the observed baryonic distribution defines the source field, one object-dependent structural acceleration scale $a_\star$ selects the nonlinear response, and the solved field is then confronted with the stellar kinematics. Rotation and pressure are therefore observational constraints on the same gravitational state, not separate gravity laws.

After the dynamical solution has been selected, CPTG also evaluates Structural Mode N , a dimensionless measure of the radial organization of the solved field. Structural Mode does not determine the field and is not a morphology fit. It is calculated afterward from the radial structure of the retained solution and may then be mapped into the Curvature Structure Mode Index (CSMI), a secondary structural catalog [9].

The analysis is organized accordingly. The observational inputs and test architecture are defined first; source units and native phase are then fixed; conventional dark-matter accounting is separated from the CPTG gravitational interpretation; the stellar source and resolved inner rotation are reconstructed; one CPTG field is solved together with the stellar pressure profile; the effective-radius rotation is compared only after that solution is fixed; and Structural Mode is evaluated last.

2 Test Architecture and Observational Basis

The calculation is designed as a resolved-inner cross-validation rather than a one-radius closure. The published effective-radius rotation is not used to determine the rotation profile or the structural acceleration scale. Instead, the inner spectrum constrains the ordered-motion profile, the observed stellar structure fixes the baryonic source, and the effective-radius dispersion supplies the pressure normalization of the declared Jeans reduction. The source value V_{R_e} is restored only after those steps are complete.

The operational sequence is

$$\begin{aligned}
 &\text{redshift and source units} \longrightarrow \text{selected native phase and converted stellar source} \\
 &\quad \quad \quad \Downarrow \\
 &\text{resolved inner rotation} \longrightarrow \{a_\star, g_{\text{CPTG}}(r), P_R(r)\} \longrightarrow \begin{cases} V_{\text{CPTG}}(R_e), \\ N \rightarrow \text{CSMI}. \end{cases}
 \end{aligned} \tag{5}$$

The quantities in this sequence have distinct roles. Redshift fixes the geometric epoch; the stellar mass and Sérsic profile define the material source; the inner velocity measurements constrain the ordered-motion profile; the intrinsic dispersion at R_e normalizes the pressure channel; and the object-specific $a_\star$ selects the CPTG field compatible with those conditions. Structural Mode is calculated only from the retained radial field.

The observational authority is the JWST-SUSPENSE analysis of Slob et al. [1]. The present resolved-inner reconstruction additionally uses the public DAWN JWST Archive G140M/F100LP two-dimensional spectrum for source 129982 [2]. Detailed file provenance, hashes, extraction evidence, and numerical closure records are listed in the Data, Code, and Evidence Availability section rather than repeated through the scientific narrative.

Measured quantity	Adopted value	Role in the test
Spectroscopic redshift	$z = 1.249$	Native phase and source-distance conversion
Effective radius	$6.48 \pm 1.13 \text{ kpc}$	Stellar-source radial scale
Sérsic index	$n = 3.6 \pm 0.1$	Stellar concentration
Projected axis ratio	$q_{\text{proj}} = 0.39 \pm 0.01$	Source projection and kinematic geometry
Slit/major-axis offset	$30 \pm 1^\circ$	NIRSpec source/slit geometry
Chabrier stellar mass	$\log_{10}(M_\star/M_\odot) = 11.22 \pm 0.01$	Baryonic normalization
Intrinsic dispersion anchor	$213^{+11}_{-2} \text{ km s}^{-1}$	Pressure normalization at R_e
Inner odd LOS velocities	$33.5, 88.0, 120.5 \text{ km s}^{-1}$	Resolved-inner rotation inference
Effective-radius rotation	$334^{+52}_{-46} \text{ km s}^{-1}$	Withheld endpoint comparison
Published dynamical mass	$\log_{10}(M_{\text{dyn}}/M_\odot) = 11.9 \pm 0.1$	Conventional-model comparator only

Table 1: Inputs and comparison quantities used in the resolved-inner CPTG test. The published effective-radius rotation is withheld from the field and rotation-profile inference.

With the roles of the observables fixed, the next step is to place the source-reported physical quantities on the CPTG comparison coordinate and identify the native phase at which the bound system is evaluated.

3 Source-Distance Convention and Native Phase

Slob et al. report physical sizes and stellar masses using a flat- Λ CDM source convention with $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [1]. These parameters define the observational reporting scale; they do not determine the CPTG native branch or evolve the galaxy’s local structural acceleration.

For consistency with the recent CPTG high-redshift structural studies, distance-dependent source quantities are re-expressed on the arithmetic midpoint of the CPTG acoustic and luminosity distance coordinates. At $z = 1.249$ the resulting scale factor is

$$s_D = \frac{D_{A,\text{mid}}}{D_A^{\Lambda 70}} = 0.99030987, \quad (6)$$

which gives

$$R_e = 6.41721 \text{ kpc}, \quad M_\star = 1.62758 \times 10^{11} M_\odot. \quad (7)$$

The conversion changes only the physical reporting coordinate; it is not a new stellar-population fit.

The observed redshift independently selects the native geometric phase,

$$a = \frac{1}{1+z} = 0.4446420631, \quad N_{\pi,\text{obs}} = \ln a = -0.810485673, \quad (8)$$

with native clock coordinate

$$t_{\pi,\text{obs}} = 4.333836 \text{ Gyr}. \quad (9)$$

Here N_π is a cosmological phase coordinate and is unrelated to the galaxy Structural Mode N introduced later. Acoustic and luminosity clock coordinates are downstream projections of the same selected phase; none is used to rescale or evolve the local $a_\star$ [8, 7].

Once the observed phase is selected, it is treated as the local present of the bound-system calculation. The gravitational comparison can therefore proceed using the measured galaxy state at that epoch.

4 Dark Matter and CPTG: Two Interpretations

The conventional and CPTG descriptions use the same observed galaxy but assign the excess gravitational response to different physical sectors. The distinction is clearest when the measured stellar kinematics are separated from the model-dependent dynamical interpretation.

For its conventional dynamical-mass estimate, Slob et al. define

$$v_{\text{circ}}^2(R_e) = V_{R_e}^2 + 3.35 \sigma_0^2, \quad (10)$$

and

$$M_{\text{dyn,tot}} = k_{\text{tot}} \frac{v_{\text{circ}}^2(R_e) R_e}{G}, \quad k_{\text{tot}} = 1.8, \quad (11)$$

for their adopted thin-disc/oblate calibration [1, 4, 5]. The central kinematics give

$$v_{\text{circ,src}} = 513.36 \text{ km s}^{-1}. \quad (12)$$

Together with Eq. (4), this produces a large dynamical mass excess over the Chabrier stellar mass. Under low-gas bookkeeping, the difference is conventionally assigned largely to a non-stellar gravitating component, principally dark matter [1].

That statement needs one important qualification. The source M_{dyn} is model-derived, not a direct measurement of a dark halo. Slob et al. explicitly identify the thin-disc and radially constant-dispersion assumptions as limitations and note that a future JAM analysis is needed to allow variable intrinsic thickness and to calibrate the stellar asymmetric-drift treatment more completely [1]. Equation (10) is therefore retained here as the source dynamical comparator, not as a universal force law.

CPTG retains the measured stellar distribution as the material source and attributes any additional local gravitational response to curvature polarization and transport. The galaxy-specific structural scale $a_\star$ is inferred from the mixed-support solution; it is neither a universal acceleration constant nor a quantity set by the cosmological Hubble rate [7].

Question	Dark-matter interpretation	CPTG calculation in this paper
Material source of the local field	Baryons embedded in a gravitating halo; the dynamical mass may exceed the directly inventoried baryonic mass	Measured stellar source; no dark-halo stress–energy component is inserted
Additional gravity source	Additional gravitating mass is available through the halo	Nonlinear curvature polarization and transport increase the geometric response to the baryonic source
Treatment of mixed stellar support	Source thin-disc model uses the effective-radius asymmetric-drift relation $V^2 + 3.35\sigma^2$ for its dynamical-mass coordinate	Ordered rotation and pressure are required to be supported by one solved CPTG field through a radial Jeans equation
Galaxy-specific structural scale	No CPTG a_* analogue	One inferred object-level a_* ; R_e is derived from it
Additional structural output	No CPTG Structural Mode analogue	Post-solution Structural Mode N , then CSMI
Scope of this paper	Source M_{dyn} provides the conventional comparison context	Local gravitational support and radial field organization are tested; galaxy-formation microphysics is not replaced

Table 2: The two interpretations use the same observed galaxy but different gravitational accounting. The source dynamical mass is a model-derived conventional comparator; it is not treated as a direct halo measurement in this paper.

The remainder of the analysis therefore proceeds from the common observational basis: the measured stellar mass distribution and the resolved mixed kinematics.

5 Reconstructing the Baryonic Galaxy

A resolved gravitational calculation requires the stellar mass to be distributed where the galaxy is actually observed. Placing the entire stellar mass inside R_e would overstate the local source field because R_e is a half-light scale and 129982 has a concentrated, strongly flattened light profile. The stellar source is therefore reconstructed from the measured Sérsic structure.

The projected surface-density profile is

$$\Sigma_*(s) = \Sigma_0 \exp \left[-b_n \left(\frac{s}{R_e} \right)^{1/n} \right], \quad (13)$$

with $n = 3.6$ and $q_{\text{proj}} = 0.39$, normalized to the converted stellar mass in Eq. (7).

For the central three-dimensional reconstruction, similar oblate density surfaces are adopted,

$$m^2 = R^2 + \frac{z^2}{q_{\text{int}}^2}, \quad (14)$$

with $q_{\text{int}} = 0.20$. This is a finite-thickness source-potential reference rather than a CPTG constant. Because projected flattening does not uniquely determine intrinsic thickness, the final Structural Mode is also evaluated over $q_{\text{int}} = 0.10\text{--}0.39$.

The exact Abel inversion of the projected elliptical profile is

$$\rho_{\star}(m) = -\frac{q_{\text{proj}}}{\pi q_{\text{int}}} \int_m^{\infty} \frac{d\Sigma_{\star}/ds}{\sqrt{s^2 - m^2}} ds, \quad (15)$$

and the equatorial acceleration of the nested oblate homoeoids is [6]

$$g_{\star,R}(R) = \frac{4\pi G q_{\text{int}}}{R} \int_0^R \frac{\rho_{\star}(m) m^2 dm}{\sqrt{R^2 - (1 - q_{\text{int}}^2) m^2}}. \quad (16)$$

The numerical deprojection reproduces the input stellar mass to approximately 9.1×10^{-6} fractionally. At the converted effective radius,

$$g_{\star}(R_e) = 2.99000 \times 10^{-10} \text{ m s}^{-2}. \quad (17)$$

For the same $n = 3.6$ profile, the projected 95% light radius used later for Structural Mode is

$$R_{95} = 7.687153 R_e = 49.33006 \text{ kpc}. \quad (18)$$

The stellar source is now fixed. The next task is to obtain the ordered-motion profile from the spectrum without using the eventual R_e comparison value.

6 Resolved Inner Rotation from the Public NIRSpec Spectrum

The ordered-motion profile is constrained directly from the public two-dimensional NIRSpec spectrum rather than from the published effective-radius endpoint. Opposite spatial rows contain a common systemic component and opposite rotational contributions; the antisymmetric combination therefore suppresses the shared zero point and isolates the resolved rotational signal. The three inner coordinates retained for the fit are

$$\hat{V}_{\text{odd}} = (33.5, 88.0, 120.5) \text{ km s}^{-1}, \quad (19)$$

with empirical weighting uncertainties of 12, 18, and 28 km s^{-1} . They extend to approximately $0.62R_e$ in the deprojected thin-disc coordinate. The source-published V_{R_e} is not included in this inference.

The intrinsic ordered rotation is represented by the same arctangent form adopted by the source analysis,

$$V_{\phi}(r) = \frac{2V_a}{\pi} \arctan\left(\frac{r}{r_t}\right). \quad (20)$$

The measured Sérsic source is projected through the observed slit geometry and an empirically calibrated residual spatial-transfer kernel. The central transfer solution gives

$$\frac{r_t}{R_e} = 0.400061, \quad V_a = 423.505 \text{ km s}^{-1}, \quad (21)$$

with $\chi_{\text{inner}}^2 = 0.2879$ for the three retained antisymmetric constraints. The fitted projected values are

$$V_{\text{odd}}^{\text{model}} = (37.51, 81.00, 124.93) \text{ km s}^{-1}. \quad (22)$$

The residual transfer calibration gives a 16th–84th-percentile range

$$0.111086'' \leq \sigma_{\text{eff}} \leq 0.117137'', \quad (23)$$

centered on $0.113766''$. This kernel represents the residual spatial response of the reduced product after the explicit source and aperture geometry are included; it is not identified with the telescope PSF alone. The detailed extraction and calibration provenance is documented in the Data, Code, and Evidence Availability section.

With the ordered-motion profile fixed from the inner data, the remaining task is to determine whether one CPTG field can support that rotation together with the measured stellar pressure.

7 Mixed-Support CPTG Field and Jeans Pressure Solution

7.1 Structural acceleration scale and nonlinear field

The object-dependent structural acceleration $a_\star$ sets the response scale of the galaxy. It is not the observed acceleration V^2/r , it is not σ^2/r , and it is not a universal constant shared by all galaxies. In the present resolved mixed-support calculation it is the single structural scale selected by requiring the same CPTG field that accompanies the resolved rotation profile to reproduce the declared stellar pressure normalization at R_e [7].

For every positive trial $a_\star$, the curvature-transport radius is derived as

$$R_c = \frac{48\pi c^2}{a_\star}, \quad (24)$$

and the radial transport remnant is

$$g_{\text{geom}}(r) = \frac{1}{2}a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \quad (25)$$

The current galaxy-limit CPTG field is the implicit equation [7]

$$g(r) = g_\star(r) + \frac{\sqrt{a_\star g_\star(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \quad (26)$$

The total acceleration is solved independently at each radius. No halo profile and no multiplicative effective-radius response factor are added.

7.2 Radial pressure boundary-value solution

The mixed-support solution requires the random stellar motions to be supported by the same radial field that governs the ordered rotation. Define the radial pressure coordinate

$$P_R(r) = \rho_\star(r) \sigma_R^2(r). \quad (27)$$

For the declared stationary reduction, the cross moment $\overline{v_R v_z}$ is neglected and the in-plane random components are taken equal, $\sigma_\phi^2 = \sigma_R^2$. Under these assumptions the radial Jeans equation becomes

$$\frac{dP_R}{dr} = \rho_\star(r) \left[\frac{V_\phi^2(r)}{r} - g_{\text{CPTG}}(r) \right]. \quad (28)$$

The physical boundary condition is

$$P_R(r) \rightarrow 0 \quad \text{as} \quad \rho_\star(r) \rightarrow 0, \quad (29)$$

so

$$P_R(r) = \int_r^\infty \rho_\star(s) \left[g_{\text{CPTG}}(s) - \frac{V_\phi^2(s)}{s} \right] ds, \quad (30)$$

and

$$\sigma_R^2(r) = \frac{P_R(r)}{\rho_\star(r)}. \quad (31)$$

The integration is extended beyond R_{95} until the tracer contribution is negligible. The boundary condition applies to pressure, not to σ_R at an arbitrarily chosen finite radius.

For the declared reduced interface, the source intrinsic dispersion parameter is used as the radial pressure anchor at the effective radius,

$$\sigma_R(R_e; a_\star) = 213 \text{ km s}^{-1}. \quad (32)$$

This gives

$$a_\star = 2.97160 \times 10^{-9} \text{ m s}^{-2}, \quad R_e = 147.806 \text{ Gpc}. \quad (33)$$

At R_e ,

$$g_{\text{CPTG}}(R_e) = 1.21987 \times 10^{-9} \text{ m s}^{-2}, \quad (34)$$

so

$$\frac{g_{\text{CPTG}}(R_e)}{g_\star(R_e)} = 4.07983. \quad (35)$$

The corresponding circular-equivalent gravitational speed is

$$\sqrt{g_{\text{CPTG}}(R_e) R_e} = 491.48 \text{ km s}^{-1}, \quad (36)$$

but it is not the ordered stellar rotation. The difference between this circular-equivalent speed and the streaming motion is carried by the solved pressure support.

The explicit transport remnant at the effective radius is only

$$g_{\text{geom}}(R_e) = 7.32 \times 10^{-20} \text{ m s}^{-2}, \quad (37)$$

so the local field enhancement is dominated by the nonlinear polarization response rather than by the explicit radial remnant.

7.3 Emergent effective pressure coefficient

At R_e , the solved state can be expressed in the familiar pressure-support form

$$g_{\text{CPTG}}(R_e)R_e = V_\phi^2(R_e) + \alpha_J \sigma_R^2(R_e), \quad (38)$$

but now the coefficient is an output,

$$\alpha_J = 3.05437. \quad (39)$$

The local three-dimensional Sérsic density slope is

$$\gamma_e = - \left. \frac{d \ln \rho_\star}{d \ln r} \right|_{R_e} = 2.743300, \quad (40)$$

and the solved dispersion-gradient contribution is

$$\delta_e = - \left. \frac{d \ln \sigma_R^2}{d \ln r} \right|_{R_e} = 0.311123. \quad (41)$$

Thus

$$\gamma_e + \delta_e = 3.054423, \quad (42)$$

which independently reproduces Eq. (39) to numerical accuracy. The source-model value 3.35 in Eq. (10) therefore remains a useful conventional comparator, but it is not used to generate the CPTG pressure profile.

r/R_e	Ordered V_ϕ (km s ⁻¹)	Radial σ_R (km s ⁻¹)
0.10	66.04	244.12
0.20	124.99	250.63
0.40	211.73	241.02
0.60	264.95	229.62
0.80	298.48	220.37
1.00	320.90	213.00
1.50	353.23	199.65
2.00	370.28	190.35
3.00	387.76	177.45
5.00	401.98	161.56
R_{95}/R_e	409.49	148.75

Table 3: Ordered-rotation and radial-dispersion profiles in the central mixed-support solution. The pressure $P_R = \rho_\star \sigma_R^2$ tends to zero with the tracer density at large radius; σ_R is not forced to vanish at R_{95} .

The field and pressure channels are now fixed without using the source-published effective-radius rotation. That value can therefore be brought back as a withheld comparison.

8 Held-Out Effective-Radius Rotation Test

After the inner rotation profile, CPTG field, and pressure normalization have been fixed, the source-published effective-radius rotation can be restored as an external comparison. The resolved-inner solution gives

$$V_{\text{CPTG,mix}}(R_e) = 320.90 \text{ km s}^{-1}, \quad (43)$$

while Slob et al. report

$$V_{R_e} = 334^{+52}_{-46} \text{ km s}^{-1}. \quad (44)$$

The central residual is

$$\Delta V = -13.10 \text{ km s}^{-1}, \quad \frac{\Delta V}{334} = -3.92\%, \quad (45)$$

placing the CPTG result within the reported 288–386 km s⁻¹ interval.

Quantity	Value	Interpretation
Observed V_{R_e}	$334^{+52}_{-46} \text{ km s}^{-1}$	Withheld source endpoint
CPTG resolved-inner result	320.90 km s^{-1}	Continuation of the inner-fitted mixed-support solution
Central residual	-13.10 km s^{-1}	-3.92% relative to the source central value
Transfer-width sensitivity	$318.85\text{--}323.82 \text{ km s}^{-1}$	Transfer sensitivity only; not a full confidence interval
Source-model v_{circ}	513.36 km s^{-1}	Conventional asymmetric-drift dynamical coordinate

Table 4: Effective-radius comparison. Stellar streaming rotation, circular-equivalent gravitational support, and the source conventional v_{circ} are distinct quantities.

Re-solving across the empirical transfer range in Eq. (23) gives $V(R_e) = 318.85\text{--}323.82 \text{ km s}^{-1}$. This span quantifies only the measured spatial-transfer sensitivity and is not presented as a complete statistical confidence interval. A formal likelihood for the intrinsic turnover and dispersion tensor would require a full replication of the source pPXF, line-spread-function, and axisymmetric dynamical analysis.

The result is therefore best described as a resolved-inner cross-validation. The effective-radius rotation is withheld, whereas the published intrinsic dispersion at R_e remains the pressure anchor used to select the object-level a_* . This is stronger than the former one-radius shared-input closure but is not a parameter-free prediction from photometry alone.

9 Structural Mode N and the CSMI Map

Structural Mode N is a dimensionless post-solution measure of the radial organization of the solved CPTG field. It is evaluated from both the field amplitude and its radial gradient, which together identify where the retained gravitational response changes most strongly. The weighted centroid of that structural variation defines a characteristic organization length λ , and the ratio $N = R/\lambda$ expresses that organization relative to the adopted outer galaxy radius [9].

Larger N therefore corresponds to structural variation weighted more strongly toward the interior, while smaller N indicates that significant variation remains distributed farther outward. Because the quantity is evaluated only after the field solution has been selected, observed Hubble type, visual morphology, spin class, and V/σ do not enter its calculation.

The registered construction begins with

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}. \quad (46)$$

The structural-change weight is

$$w(r) = \left| \frac{dC_g}{dr} \right|, \quad (47)$$

and its radial centroid is

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}. \quad (48)$$

Structural Mode is then

$$N = \frac{R}{\lambda}, \quad R = R_{95}. \quad (49)$$

The Curvature Structure Mode Index (CSMI) is a downstream catalog map applied only after the continuous value of N has been calculated. Its labels describe intervals of CPTG field organization and should not be interpreted as direct visual morphology assignments.

Structural Mode N	Current CSMI Type
$N \leq 1.45$	Dwarf Irregular
$1.45 < N \leq 1.75$	Magellanic Irregular
$1.75 < N \leq 1.95$	LSB Dwarf Disk
$1.95 < N \leq 2.15$	Transition Dwarf
$2.15 < N \leq 2.35$	LSB Spiral
$2.35 < N \leq 2.55$	Very Late Spiral
$2.55 < N \leq 2.80$	Late Spiral
$2.80 < N \leq 3.05$	Intermediate Spiral
$3.05 < N \leq 3.30$	Early Spiral
$3.30 < N \leq 3.55$	Bulged Spiral
$3.55 < N \leq 3.72$	Lenticular/Early Disk
$N > 3.72$	High-Mode Outlier

Table 5: Current CSMI map associated with the validated galaxy Structural Mode implementation. The continuous value of N is the primary scientific quantity; the CSMI label is secondary.

For the central $q_{\text{int}} = 0.20$ field,

$$\lambda = 16.89915 \text{ kpc}, \quad N_{\text{obl}} = 2.91909. \quad (50)$$

This lies in the current Intermediate-Spiral interval, $2.80 < N \leq 3.05$. The CSMI designation is structural only; the observational description of 129982 remains that of a massive quiescent fast rotator.

Intrinsic thickness is not uniquely measured, so the source potential is re-solved over a finite geometry family.

q_{int}	$a_{\star} \text{ (m s}^{-2}\text{)}$	N	CSMI readout
0.10	2.8100×10^{-9}	2.93287	Intermediate Spiral
0.20	2.9716×10^{-9}	2.91909	Intermediate Spiral
0.30	3.1306×10^{-9}	2.90639	Intermediate Spiral
0.39	3.2720×10^{-9}	2.89577	Intermediate Spiral

Table 6: Intrinsic-source geometry sensitivity. The kinematic thin-disc projection is held fixed while the oblate source potential, $a_{\star}$, CPTG field, and Structural Mode are re-solved. The continuous mode remains in the same CSMI interval throughout the tested source-thickness family.

The geometry test therefore shows that the post-solution structural placement is stable across the adopted intrinsic-thickness range, independently of the effective-radius velocity comparison.

10 Discussion and Limitations

The resolved-inner calculation returns a coherent mixed-support solution with one inferred structural acceleration scale. Table 7 summarizes the principal numerical results and their roles in the analysis.

Final quantity	Central result	Scientific role
Structural acceleration $a_\star$	$2.9716 \times 10^{-9} \text{ m s}^{-2}$	Galaxy-specific CPTG response scale
Transport radius R_c	147.806 Gpc	Derived from $a_\star$
Stellar field $g_\star(R_e)$	$2.9900 \times 10^{-10} \text{ m s}^{-2}$	Reconstructed baryonic source
CPTG field $g_{\text{CPTG}}(R_e)$	$1.21987 \times 10^{-9} \text{ m s}^{-2}$	Solved geometric response
Emergent pressure coefficient α_J	3.05437	Density plus solved dispersion-gradient support
Held-out $V_{\text{CPTG}}(R_e)$	320.90 km s^{-1}	Resolved-inner cross-validation result
Observed V_{R_e}	$334^{+52}_{-46} \text{ km s}^{-1}$	Source endpoint comparator
Structural Mode N	2.91909	Post-solution radial organization
CSMI readout	Intermediate Spiral	Downstream structural catalog label

Table 7: Central results of the resolved-inner mixed-support calculation. The field response, kinematic continuation, and Structural Mode are distinct outputs of the same retained solution.

The main gravitational result is the simultaneous support of substantial ordered rotation and stellar pressure without inserting the source-inferred dark-halo mass into the CPTG field equation. The central CPTG field at R_e exceeds the direct stellar-source field by a factor of 4.07983, while the corresponding mixed-support continuation gives $V(R_e) = 320.90 \text{ km s}^{-1}$ against the source value $334^{+52}_{-46} \text{ km s}^{-1}$. In the conventional source interpretation, the same galaxy instead carries a large dynamical-to-stellar mass excess under the adopted thin-disc asymmetric-drift model. The two descriptions therefore differ in where the additional gravitational response is assigned, not in the underlying observed stellar data.

The present result does not remove the need for ordinary galaxy astrophysics. CPTG does not replace stellar-population evolution, gas dynamics, quenching, mergers, black-hole growth, or feedback. It addresses the gravitational accounting of the already-observed bound system. Likewise, the source dynamical mass is a model-derived quantity and is not treated here as a direct measurement of a dark halo.

Several limitations remain material to the interpretation. The Jeans reduction retains one radial random-motion channel rather than a complete three-integral distribution function or full JAM reconstruction with independently constrained σ_R , σ_ϕ , σ_z , tilt, and intrinsic thickness. The public reduced spectrum constrains the ordered-motion gradient more directly than the full stellar dispersion tensor; consequently, the identification $\sigma_R(R_e) = 213 \text{ km s}^{-1}$ is a declared pressure-interface assumption. The empirical transfer kernel is calibrated from the reduced continuum profile and should not be identified with the telescope PSF alone. Its quoted sensitivity range is not a full statistical confidence interval. The stellar-source reconstruction assumes constant M/L , and the Structural Mode calculation extends the fitted Sérsic source through the faint outer region to R_{95} .

These limitations define a clear refinement path rather than an alternative interpretation of the present calculation. A full pPXF/LSF/JAM replication can test the orbital decomposition and tighten the statistical uncertainty while preserving the same inference hierarchy: measured stellar source, one CPTG field, simultaneous ordered and random support, withheld kinematic comparison, and post-solution Structural Mode.

11 Conclusion

JWST-SUSPENSE 129982 provides a resolved CPTG test in a regime where strong ordered stellar rotation and substantial random motion coexist in a massive quiescent galaxy at cosmic noon. The system therefore supports a direct comparison between conventional dark-matter dynamical accounting and CPTG geometric-response accounting.

Using the public resolved spectrum only inside approximately $0.62R_e$, the analysis determines the inner rotation profile while withholding the source-published effective-radius rotation. The oblate Sérsic stellar source, nonlinear CPTG field, and radial Jeans pressure solution then select

$$a_\star = 2.9716 \times 10^{-9} \text{ m s}^{-2}, \quad V_{\text{CPTG,mix}}(R_e) = 320.90 \text{ km s}^{-1}. \quad (51)$$

The withheld observational value is $334_{-46}^{+52} \text{ km s}^{-1}$, corresponding to a central residual of -13.10 km s^{-1} or -3.92% . The effective pressure coefficient emerges from the solved state as $\alpha_J = 3.05437$ rather than being imposed from the source’s conventional 3.35 asymmetric-drift prescription.

The same retained field gives Structural Mode $N = 2.91909$, which falls in the current Intermediate-Spiral CSMI interval. This classification refers to the radial organization of the solved CPTG field and is independent of the galaxy’s observed fast-rotator designation. Within the declared reduction, one baryon-sourced CPTG field therefore supports the resolved mixed stellar dynamics and yields a stable post-solution structural coordinate without inserting a dark-halo gravitational source.

Data, Code, and Evidence Availability

The primary observational authority is Slob et al. [1]. The JWST-SUSPENSE observations are associated with JWST program GO-2110 and MAST DOI 10.17909/y6rb-fn24. The public reduced G140M/F100LP spectrum used for the resolved-inner reconstruction is distributed through the DAWN JWST Archive [2].

The exact spectrum used in this manuscript is:

`suspense-kriek-v4_g140m-f100lp_2110_129982.spec.fits`

SHA-256: 190c9e9953fdd7237459f2ab20d01e847c6940efe75d4890a75816b449764b70

The numerical evidence record documents positive-trace selection, removal of the negative nodded replicas, the retained antisymmetric velocity coordinates and weighting uncertainties, the eleven-window spatial-transfer calibration, the nonlinear field solve, the radial Jeans integration, geometry sensitivity, and the Structural Mode calculation. The closure record used for the manuscript values is:

`CPTG_SUSPENSE_129982_ResolvedInner_Closure_20260902_r06_REPORT.txt`

SHA-256: fee644d6b74b0093f1d86bb762c70890402e1c41614d31b4daf3ddcc685a6be

The source reconstruction, resolved-inner velocity inference, spatial-transfer calibration, nonlinear CPTG field solve, radial Jeans pressure integration, and Structural Mode calculation are deterministic numerical operations. Numerical calculations retain additional guard digits; manuscript values are rounded for interpretation. Supporting CPTG theory and numerical resources are maintained at <https://github.com/CLG2025/CPTG>.

References

- [1] M. Slob et al., *Fast rotators at cosmic noon: Stellar kinematics for 15 quiescent galaxies from JWST-SUSPENSE*, *Astron. Astrophys.* **702**, A110 (2025), DOI: 10.1051/0004-6361/202555812.
- [2] DAWN JWST Archive, *Public JWST/NIRSpec reduced spectra and extraction products*, <https://dawn-cph.github.io/dja/>.
- [3] G. Chabrier, *Galactic Stellar and Substellar Initial Mass Function*, *Publ. Astron. Soc. Pac.* **115**, 763–795 (2003).
- [4] A. Burkert, R. Genzel, N. Bouché, et al., *Astrophys. J.* **725**, 2324 (2010).
- [5] S. H. Price, H. Übler, N. M. Förster Schreiber, et al., *Astron. Astrophys.* **665**, A159 (2022).
- [6] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd ed., Princeton University Press (2008).
- [7] Carter L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, 2026, DOI: 10.5281/zenodo.22441039.
- [8] Carter L. Glass Jr., *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates*, 2026, DOI: 10.5281/zenodo.22426857.
- [9] Carter L. Glass Jr., *Structural Mode N in CPTG: Galaxy Structural Classification from the Solved CPTG Field*, 2026, DOI: 10.5281/zenodo.22436682.